STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a Brooklyn Union of L.I. for Gas Service	**Case 19-G-0310**

Prepared Direct Testimony of:

New York City Rate Design Panel:

Richard Baudino
Sergey Shabalin

On Behalf of:

The City of New York

August 30, 2019

INTRODUCTION

Q. PLEASE STATE YOUR NAME AND BUSINESS ADDRESS.

A. Mr. Baudino – My name is Richard A. Baudino. My business address is J. Kennedy and Associates, Inc. ("Kennedy and Associates"), 570 Colonial Park Drive, Suite 305, Roswell, Georgia 30075.

Mr. Shabalin – My name is Sergey Shabalin. My business address is DCAS Energy Management, David N. Dinkins Municipal Building, 1 Centre Street, 17th floor, New York, NY 10007.

Q. WHAT IS YOUR OCCUPATION AND BY WHOM ARE YOU EMPLOYED?

A. Mr. Baudino – I am a consultant to Kennedy and Associates.

Mr. Shabalin – I am a Director, Load Management and Utility Affairs for City of New York, Department of Citywide Administrative Services, Division of Energy Management ("DCAS DEM").

Q. PLEASE DESCRIBE YOUR EDUCATION AND PROFESSIONAL EXPERIENCE.

A. Mr. Baudino – I received my Master of Arts degree with a major in Economics and a minor in Statistics from New Mexico State University in 1982. I also received my Bachelor of Arts Degree with majors in Economics and English from New Mexico State in 1979. I began my professional career with the New Mexico Public Service Commission Staff in October 1982 and was employed there as a Utility Economist. During my employment with the Staff, my responsibilities included the analysis of a broad range of issues in the ratemaking field. Areas in which I testified included cost of service, rate of return, rate

design, revenue requirements, analysis of sale/leasebacks of generating plants, utility finance issues, and generating plant phase-ins.

In October 1989, I joined the utility consulting firm of Kennedy and Associates as a Senior Consultant where my duties and responsibilities covered substantially the same areas as those during my tenure with the New Mexico Public Service Commission Staff. I became Manager in July 1992 and was named Director of Consulting in January 1995. Currently, I am a consultant with Kennedy and Associates. Exhibit ____(NYCRDP-1) summarizes my expert testimony experience.

Mr. Shabalin – I received my Master of Science degree with a major in Energy Management from New York Institute of Technology in 2012. I also received my Bachelor of Arts degree with a major in Chemistry from New York University in 2001. I joined DCAS DEM in 2012 as an Energy Billing Analyst focusing on utility billing and tariff matters. I became Director of Billing and Reporting Analytics in 2015. My current role at DEM includes overseeing programs that leverage data-backed operational capabilities of municipal facilities to contribute to the integrity of the City's energy supply (such as demand response and interruptible gas), derive energy and carbon savings (load management) and deployment of real-time monitoring technology across the targeted municipal portfolio. Prior to joining DEM in 2012, I spent time at a Long Island City, New York non-profit Community Environmental Center working on energy audits and energy retrofits in New York's affordable housing stock.

Q. ON WHOSE BEHALF ARE YOU TESTIFYING?

A. We are testifying on behalf of the City of New York.

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. The purpose of our testimony is to address cost and revenue allocation, rate design, and service quality issues for Brooklyn Union Gas Company ("KEDNY") and KeySpan Gas East Corporation ("KEDLI"). In so doing, we will address the Direct Testimony of and certain recommendations made by the Companies' Rate Design Panel ("RDP").

Q. PLEASE SUMMARIZE YOUR CONCLUSIONS AND RECOMMENDATIONS TO THE PUBLIC SERVICE COMMISSION OF THE STATE OF NEW YORK ("COMMISSION").

A. Mr. Baudino: my conclusions and recommendations with respect to cost and revenue allocation and rate design are set forth below. Except where specifically noted, my recommendations relative to KEDNY also apply to KEDLI.

1. The embedded cost of service studies ("ECOSS") that the Companies filed are reasonable, consistent with past practice, and should be accepted by the Commission as the primary guide for revenue allocation.
2. With the exceptions noted in the next bullet regarding the KEDLI TC/IT class, The RDP's proposed interclass revenue allocations for KEDNY and KEDLI appear to be reasonable given the ECOSS results.
3. Although the KEDNY TC/IT class could have its rates and revenue reduced even further, I support the proposed small reduction to that class, as set forth by the KEDNY RDP in its Corrections and Update ("C&U Filing"). Under no circumstances should the KEDNY TC/IT classes receive any rate increase in this

proceeding. With respect to KEDLI, I do not agree with the RDP's proposed increases for the TC/IT service class. This class should receive no increases in this case given its excessive current rate of return.

4. The Commission should approve the RDP's proposed restructuring of the TC/IT classes into two new Non-firm rate classes with Tier 1 and Tier 2 interruption levels. As noted earlier, the KEDNY rates for the new Service Classes ("SC") 18-22 and 22, as set forth in the C&U Filing, should be approved. The KEDLI rates for new Service Classes 18 and 19, as set forth in the C&U Filing also should be set so that these classes receive no increase.

5. The Commission should order the Companies to modify their proposed new Non-firm tariffs to specifically state the interruption temperatures as 15 degrees for Tier 1 and 20 degrees Fahrenheit for Tier 2.

6. In connection with the Companies' proposed new Non-firm service class, customers taking service under those new Non-firm rate schedules and have automatic equipment should be able to elect to take service under Tier 2.

7. Exposing Non-firm customers to System Benefits Charges ("SBC") should be deferred until adequate energy efficiency programs are available to those customers.

8. In light of continuing capacity problems, the Companies should expand the Demand Response Demonstration Project aggressively, beyond what the Companies have proposed.

9. The Non-firm tariffs should be modified so that on-site alternative fuel is not an absolute prerequisite for eligibility.

In addition to the recommendations sponsored primarily by Mr. Baudino, we recommend the following:

10. To the extent that they are not addressed in Cases 16-G-0058 and 16-G-0059, the Commission should adopt the positions advocated by the City in its Comments filed in those dockets, which are attached hereto.

11. The second mandatory test for Non-firm customers should be eliminated.

12. The monetary penalty for so-called "second strike" violations should be reduced or waived if the second interruption notice is issued within five days of the end of a prior event *and* the customer can demonstrate that it could not repair the equipment that prevented it from interrupting. In addition, under these limited circumstances, the violation should not count as a second "strike."

13. Under certain circumstances, monetary penalties otherwise applicable to the New York City Housing Authority for using gas to operate on-site generation facilities to provide electricity to human needs customers during electric emergencies should be waived.

COST AND REVENUE ALLOCATION

Q. HAVE YOU REVIEWED THE ECOSS STUDY PRESENTED BY THE COMPANIES?

A. Yes. The Companies' ECOSS and proposed revenue allocation was presented by the Rate Design Panel. The Company's ECOSS is a traditional study wherein revenue requirements are allocated among customer classes on the basis of cost causation.

Q. WHY IS AN ECOSS IMPORTANT?

A. An ECOSS illustrates the costs a utility incurs to serve each customer class. It is widely accepted that costs should be allocated among customer classes on the basis of cost causation. That principle is perhaps the most universally accepted tenet of allocating costs that cannot be directly assigned to a particular customer class. As such, costs should be allocated to those classes on the basis of how or why those costs are incurred by the utility. The results of such studies are normally used in assigning cost and revenue responsibilities to various customer classes.

Q. DO YOU SUPPORT THE PREMISE THAT COST CAUSATION PRINCIPLES SHOULD GUIDE THE ALLOCATION OF COSTS TO THE CUSTOMER CLASSES?

A. Yes. Rates that are based on consistently applied cost causation principles are fair and reasonable and further the cause of stability, conservation and economic efficiency. Other factors such as simplicity, gradualism, economic development and ease of administration may also be taken into consideration when determining the final allocation of the revenue requirement among classes, but the fundamental starting point and guideline should be the cost of serving each customer class produced by the ECOSS.

Q. PLEASE EXPLAIN THE PURPOSE AND DEVELOPMENT OF A CLASS COST OF SERVICE STUDY.

A. A class cost of service study allocates and assigns the total cost of providing utility service to the classes of customers receiving that service. The development of a class cost of service study consists of three steps: functionalization, classification, and allocation. Functionalization refers to the process by which the Company's investments and expenses are identified and segregated into different cost categories. For KEDNY and KEDLI, these functional categories include supply and storage, transmission, distribution, metering and billing, collections, sales promotions and procurement.

Once functionalization is complete, the utility's costs are classified into categories of demand-related, energy related and customer-related costs in order to facilitate the allocation of costs by applying cost causation principles. In general, all distribution costs are either responsive to increases in natural gas demand or to the number of customers. Demand-related costs are those that are needed to serve the winter peak demands of distribution customers. Demand-related costs are fixed and do not vary with the monthly and annual commodity consumption of the utility's customers. Customer-related costs are associated with the number of customers and are costs that are incurred to connect customers to the system independent of the customer's demand and energy requirements. Primary examples of customer-related costs are investments in meters, services, and a portion of main investment incurred to extend the distribution system to the customer's premises and conform to local and national codes and standards. In addition, such accounting functions as meter reading, bill preparation and revenue accounting are generally considered customer-related costs.

Lastly, the functionalized and classified costs are allocated to the various customer classes based on each class' contribution to the respective cost classifications. In general, demand-related costs are allocated based on each class' contribution to the maximum demand placed on the system by the classes or by the customers within the classes. Customer related costs are allocated based upon the number of customers in each class, weighted to account for the complexity of servicing the needs of the different classes of customers.

Q. HOW DID THE COMPANIES' ECOSS CLASSIFY DISTRIBUTION MAINS?

A. The RDP explained on pages 17 and 18 of its Direct Testimony that it used a minimum-system study to classify distribution mains into customer and demand related costs. Based on this study, 47.08% of the distribution system was classified as customer-related and 52.92% was classified as demand-related. The customer portion of distribution mains was allocated to customer classes based on the number of customers and the demand component was allocated based on class contribution to design-day demand.

Q. IS THE COMPANIES' ALLOCATION OF DISTRIBUTION MAIN COSTS ON A CUSTOMER AND DEMAND BASIS CONSISTENT WITH THE NATIONAL ASSOCIATION OF REGULATORY UTILITY COMMISSIONERS ("NARUC") GAS DISTRIBUTION RATE DESIGN MANUAL DATED JUNE, 1989 ("1989 NARUC MANUAL")?

A. Yes. The NARUC Manuals discuss a number of methodologies and approaches to cost allocation. With respect to the allocation of distribution main costs, the 1989 NARUC Manual states that "[a] portion of the costs associated with the distribution system may be

included as customer cost." (1989 NARUC Manual at 22.) The 1989 NARUC Manual states further that, "[o]ne argument for inclusion of distribution related items in the customer cost classification is the zero or minimum size theory." (*Id*.)

Q. HAS THE COMMISSION RECOGNIZED THE APPROPRIATENESS OF A CUSTOMER COMPONENT IN THE COST OF GAS DISTRIBUTION MAINS?

A. Yes. The Commission has accepted the use of a minimum-system methodology to classify and allocate distribution costs in past National Grid cases.

Q. DO YOU AGREE WITH THE COMPANIES' CLASSIFICATION AND ALLOCATION OF DISTRIBUTION MAINS?

A. Yes. As the RDP pointed out, distribution mains are installed to connect customers to the distribution system and to provide capacity to meet the winter peak. The Companies' ECOSS appropriately utilized a minimum-system study to estimate which portions of distribution mains are demand related and customer related, and then allocated costs to the various customer classes based on that relationship. We recommend that the Commission adopt the Companies' minimum-system methodology for classifying and allocating the cost of distribution mains.

REVENUE ALLOCATION AND RATE DESIGN

Q. WHAT WERE THE RESULTS OF THE ECOSS STUDY AS FILED BY KEDNY?

A. Table 1 below summarizes the class rate of return results of the ECOSS for KEDNY. Table 1 shows the current class rates of return and the relative rates of return at current rates. The class RORs and relative RORs presented in Table 1 were taken from Exhibit ___(RDP-4CU), Schedule 1 filed by the RDP as part of the C&U Filing made on July 3, 2019.

Table 1

KEDNY
Current Class Rates of Return

	Return %	Relative ROR %
SC1A - Res NonHeat	-0.98%	(0.29)
SC-1B Res Heat	4.23%	1.24
SC-1DG Res DG	-6.92%	(2.03)
SC-2-1 Non-Res NonHeat	9.39%	2.75
SC-2-2 Non-Res Heat	2.04%	0.60
SC-3 Multiple Family	1.76%	0.52
SC-4A High Load Factor	11.85%	3.48
SC-4A CNG	6.49%	1.90
SC-4B Year Round AC	16.25%	4.77
SC-6C TC/IT	29.13%	8.54
SC-7 Seasonal	-17.70%	(5.19)
SC-21 DG Sales	-14.30%	(4.19)
Total	3.41%	1.00

Table 1 shows that SC-1A, SC-1DG, SC-7, and SC-21 are actually earning negative returns currently. SC-2-1, SC-4A High Load Factor, SC-6C TC/IT are significantly overearning, which means that the rates these classes are currently paying are significantly greater than the cost based rates that they should be paying.

Q. HOW DID KEDNY PROPOSE TO ALLOCATE ITS REQUESTED REVENUE INCREASE IN THIS PROCEEDING?

A. Table 2 presents the RDP's recommended percentage revenue increases by service class, again from Exhibit ___(RDP-4CU), Schedule 1. This Table also presents the percentage revenue increase and decreases that would be necessary to bring each service class to the overall system average return of 6.76%.

Table 2

KEDNY
Distribution Revenue Allocation

Schedule	Full ECOSS Increase	Company Propsed Incr.
SC1A - Res NonHeat	44.4%	18.6%
SC-1B Res Heat	9.7%	13.1%
SC-1DG Res DG	156.2%	18.6%
SC-2-1 Non-Res NonHeat	-11.7%	3.3%
SC-2-2 Non-Res Heat	24.4%	18.6%
SC-3 Multiple Family	22.6%	18.6%
SC-4A High Load Factor	-16.9%	3.3%
SC-4A CNG	-0.6%	3.3%
SC-4B Year Round AC	-30.3%	3.3%
SC-6C TC/IT	-54.9%	-2.5%
SC-7 Seasonal	146.7%	18.6%
SC-21 DG Sales	312.1%	18.6%
Total	13.8%	13.8%

The RDP's Direct Testimony, pages 26 and 27, presented the methodology it used to assign the revenue increase among KEDNY's service classes. A tolerance band was developed that used a rate of return that is 0.75 times the average at the low end of the band and 1.25 times the average at the high end of the band. The service classes that were within the tolerance band were SC 1B and SC 2-2. Service classes that had class returns above the upper end of the tolerance band were assigned revenue targets five percent over proforma revenue at present rates, with the exception of TC/IT, which was assigned a 2% increase in the Companies' Direct Testimony. Service classes below the lower end of the tolerance band were assigned revenue targets of 28.2% over proforma revenue at present rates.

In the Corrections and Updates Testimony filed by the KEDNY RDP, it explained that in order to update the service class revenue allocation, the revenue allocation from the Direct Testimony was multiplied by 65.8594%, with the exception of the SC 22 TC/IT

service class, which received a 2.5% decrease in order to produce rates no higher than the April 30, 2019 filing.

Q. WHAT IS YOUR CONCLUSION WITH RESPECT TO THE PROPOSED INTERCLASS REVENUE ALLOCATION PROPOSED BY KEDNY IN THIS PROCEEDING?

A. Generally, the revenue allocation proposed by the KEDNY RDP appears reasonable given the ECOSS results and tends to move the service classes toward their cost of service while observing the ratemaking principles of gradualism and avoiding rate shock. The class increases are limited to 1.35 times the system average increase for those classes below their allocated cost of service. For purposes of this case, we support the RDP's proposed rates and revenue allocation for the new SC 18-22 and SC-22 TC/IT classes, although TC/IT customers are providing a very large subsidy to other service classes and could have their rates reduced more significantly based on the results of the Companies' ECOSS.

Q. DID YOU REVIEW KEDLI'S PROPOSED INTERCLASS REVENUE ALLOCATION FOR ITS SERVICE CLASSES?

A. Yes. We reviewed the RDP's proposed revenue allocation for KEDLI. Table 3 below presents the KEDLI distribution revenue allocation in a similar fashion to Table 2 above, with full cost based class rate increases and the RDP's proposed service class increases.

TABLE 3

KEDLI
Distribution Revenue Allocation

Schedule	Full ECOSS Increase	Company Propsed Incr.
SC1A - Res NonHeat	47.9%	12.5%
SC-1B Res Heat	10.6%	6.4%
SC-1DG Res DG	204.6%	12.5%
SC-2-A GS NonHeat	-44.1%	3.9%
SC-2-B GS Heat	5.6%	6.1%
SC-3A Multi Family	-25.1%	3.9%
SC-3B Multi-Family Heat	4.3%	6.1%
SC-9 Wholesale NGV	-81.6%	-36.0%
SC-4-7-12-13 TC/IT	-61.6%	3.9%
SC-15 HLF Service	-21.0%	3.9%
SC-16 Year Round Space Htg.	-30.6%	6.1%
SC-17 DG Sales	26.1%	12.5%
Total	6.2%	6.2%

Note that the TC/IT class receives a 3.9% increase under the RDP proposed revenue allocation despite needing a revenue decrease of 61.6% to achieve cost based rates. Given how far above costs the KEDLI TC/IT customers are based on the ECOSS, and given that these customers could have their rates significantly reduced, we recommend no revenue increase for KEDLI's TC/IT's customers in this case.

Q. ARE YOU RECOMMENDING THAT THE COMMISSION ADOPT THE COMPANIES' PROPOSED REVENUE REQUIREMENTS?

A. No. We have not evaluated the reasonableness of the Companies' proposed revenue requirements.

Q. IF THE COMMISSION APPROVES A LOWER REVENUE REQUIREMENT INCREASE THAN KEDNY OR KEDLI ARE SEEKING IN THIS CASE, HOW SHOULD THIS LOWER INCREASE BE SPREAD AMONG THE COMPANIES' SERVICE CLASSES?

A. If the Commission approves a lower revenue increase for the Companies, then we recommend that the service class increases shown in Table 2 (KEDNY) and Table 3 (KEDLI) be scaled back proportionately, with the exception of the KEDLI TC/IT class, which should remain at zero increase.

Q. PLEASE SUMMARIZE THE RDP'S PROPOSED CHANGES FOR THE TC/IT CLASSES.

A. The KEDNY RDP explained its proposal regarding TC/IT customers beginning on page 32 of its Direct Testimony. Based on the Commission's "Order Approving Tariff Revisions and Directing Further Tariff Filings" issued on February 7, 2019 ("TC/IT Order") in Cases 16-G-0058 and 16-G-0059, the RDP proposed to implement a single service classification (SC 22) with two tiers that includes both TC and IT customers. Tier 1 has a Designated Interruption Temperature of 15 degrees Fahrenheit and Tier 2 has a Designated Interruption Temperature of 20 degrees Fahrenheit. The proposed minimum monthly charge for all SC 22 customers is $375.00. In its C&U Filing, the proposed volumetric per therm rates are $0.3174 for Tier 1 customers and $0.3015 for Tier 2 customers. Tier 2 customers receive a lower rate because, at an interruption temperature of 20 degrees Fahrenheit, they will be interrupted more frequently than Tier 1 customers.

Tier 1 customers would have fully automatic switchover equipment installed so that the Company would be able to remote switch over the customers' equipment to an alternate fuel in the event it calls an interruption. Tier 2 customers would be notified of an interruption through the Company remotely activating an alarm at the semi-automatic Customer sites that would notify customers to manually switch over to their alternate fuel. As written, customers who have automatic switching equipment are limited by KEDNY and KEDLI to Tier 1 service.

Q. DO YOU AGREE WITH THIS LIMITATION?

A. No. Customers with automatic switching equipment are the most reliable source of interruptible capacity, and if they are willing to accept more interruptions in exchange for a lower rate, it is in the Companies' best interest to afford them that option.

Q. ARE THE COMPANIES WILLING TO OFFER THAT OPTION?

A. Apparently, yes. In their response to CNY-54, the Companies stated that Non-Firm customers with automatic switching equipment will automatically be placed into Tier 1, but that if a customer wishes to switch from Tier 1 to Tier 2, it can do so if it formally makes such an election. Please refer to Exhibit ___(NYCRDP-2) for the Company's response to CNY-54.

The Companies did not state in their response how long a Non-firm customer would have to make a formal election to switch to Tier 2 service when new rates are approved in this case. We recommend that customers be given 120 days from Commission approval of the new tariffs to make the election to switch from Tier 1 to Tier 2.

We also recommend that the new TC/IT tariff for both KEDNY and KEDLI include an ongoing provision to allow a customer to switch back from Tier 2 to Tier 1 no later than 120 days prior to the winter heating season. This provision would allow for Tier 2 customers who made the switch from Tier 1 to switch back in the event those customers realize that Tier 2 has too many interruptions. This is a reasonable ongoing provision that would allow TC/IT customers more flexibility and give the Companies adequate prior notice of switching before the heating season begins.

We also recommend that, within 30 days of the Commission's approval of the new tariff, the Companies provide written notice of this option to all customers that otherwise would default into the Tier 1 rate.

Q. DID YOU REVIEW THE COMMISSION'S TC/IT ORDER?

A. Yes.

Q. DID THE COMPANIES FILE PROPOSED TARIFFS FOR TC/IT CUSTOMERS CONSISTENT WITH THE TC/IT ORDER?

A. Subject to our prior comment regarding the option to elect Tier 2 service, yes. The Commission's TC/IT Order ordered the Companies to:

(1) File tariff amendments to modify the penalties for the unauthorized use of gas to the lower of two times the sum of the market price plus the transportation rate, or nine times the applicable sales rate.

(2) File proposed tariff revisions to create and/or revise their current Non-firm service classes to blend the existing TC and IT services and implement two pricing tiers determined by a customer's switching equipment.

(3) File proposed tariff revisions that address the submitted written comments and concerns for the treatment of critical customers in Case 18-G-0565, and Staff's proposed solutions discussed at the IT service technical conference on November 27, 2018.

(4) End the existing moratorium on new TC customers to become effective March 1, 2019.

The Companies' proposed Non-firm tariffs include the ordered language modifying the penalties for the unauthorized use of gas and blends the existing TC/IT classes into one Non-firm class with two pricing tiers based on a customer's switching equipment. The Companies also ended the moratorium on new Non-firm customers. Nevertheless, although the Companies generally complied with the TC/IT Order, in this testimony and in Comments filed in Cases 16-G-058 and 16-G-0059, the City is recommending additional changes to improve the Non-firm service option.

Q. TURNING TO RATE DESIGN, PLEASE COMMENT ON THE RDP'S PROPOSED VOLUMETRIC RATES FOR TIER 1 AND TIER 2 CUSTOMERS FOR KEDNY.

A. The Companies' proposed volumetric rates for Non-firm customers, as set forth in the C&U Filing, are discounted below the corresponding firm service class tail block rates to recognize the lower costs that such customers impose on the system. As noted earlier, based on the ECOSS results, these discounts should be much larger, and the volumetric rates much lower, than the Companies proposed in their C&U Filings. However, given the level of revenue increase in the Companies' filings, we do not oppose the proposed C&U rates

at this time. However, the Commission certainly should not increase Non-firm rates in this proceeding given the huge subsidy being provided by these customers.

Q. ARE TC/IT CUSTOMERS IMPORTANT IN TERMS OF HELPING KEDNY AND KEDLI MANAGE THEIR SYSTEMS DURING PEAK WINTER PERIODS?

A. Yes, and the Commission recognized this fact in its TC/IT Order, page 2, as follows:

> "By offering IT and TC service, the Companies can provide non-firm customers with delivery service during the vast majority of the year, while also ensuring the reliability of the distribution networks when demand for natural gas is highest, generally during cold weather when firm customers use natural gas for heating. Additionally, the Companies design, construct and maintain their distribution networks to serve firm customers when the distribution system experiences peak demand. Thus, the distribution networks do not include the additional infrastructure that would be required to serve TC and IT customers when the system experiences peak demand. This minimizes the cost of the system that must be recovered from firm customers."

The Commission also stated the following on page 14 of the TC/IT Order:

> "Since the Rate Order, it has become apparent that the downstate region is facing a significant and increasing shortage of firm gas transportation capacity. On November 14, 2018, National Grid filed proposed tariff amendments that are intended to provide the Companies flexibility in accepting applications for new or additional gas service if they determine there is insufficient gas supply and/or capacity to serve the prospective customers in the relevant area of their service territories. While the capacity constraints are more acute on Con Edison's gas distribution network, National Grid's load is also growing, which underscores the importance of continuing interruptible services at reasonable terms."

TC/IT customers are an invaluable part of National Grid's system. These customers are even more important now that National Grid is facing serious system constraints during the peak winter season. Continuing to offer TC/IT service at attractive, cost-based rates will be an important part of National Grid's future system planning and will offer the opportunity for the Companies to offer firm service to existing and new customers during a time of constrained capacity.

Q. ON PAGE 57 OF THE KEDNY RDP'S DIRECT TESTIMONY, THE RDP TESTIFIED THAT INTERRUPTIONS WOULD OCCUR FOR TIER 1 AND TIER 2 SC 22 CUSTOMERS AT 15 DEGREES FAHRENHEIT AND 20 DEGREES FAHRENHEIT, RESPECTIVELY. IS THIS CONSISTENT WITH KEDNY'S PROPOSED SC 22 TARIFF?

A. No. Under the Special Provisions section of the proposed SC 22 tariff, the Designated Temperatures are defined as follows:

> "The "Designated Temperature(s)" will be the temperatures set annually prior to the winter season by the Company for each tier. The Company reserves the right to adjust this temperature setting, based on system needs, with a minimum forty-eight (48) hours' notice to Customers. There will be a "Designated Interruption Temperature" for each tier when a Customer's equipment will be switched over to an alternate fuel and a "Designated Resumption Temperature" for each tier when the Customer's equipment may be switched back to natural gas."

This language is similar to the language that defines Designated Temperatures in the Companies' current TC/IT tariffs. It is not consistent with the KEDNY RDP's testimony that explicitly states the interruption temperatures. The KEDLI tariffs have the same problem. Therefore, we recommend that the Commission order the Companies to modify the language in the applicable tariff Special Provisions to explicitly state that the Designated Interruption Temperature and Designated Resumption Temperature are 15 degrees Fahrenheit for Tier 1 and 20 degrees Fahrenheit for Tier 2.

Q. DID THE COMPANIES ADDRESS THE STAFF'S RECOMMENDATIONS IN CASE 18-G-0585 REGARDING THE AVAILABILITY OF ENERGY EFFICIENCY PROGRAMS TO TC/IT CUSTOMERS?

A. Yes, in its KEDNY Direct Testimony on pages 58 and 59, the RDP stated that its proposed tariffs will have all interruptible gas service customers contribute to and be eligible for energy efficiency programs. The RDP's proposal would have all Non-firm customers contribute to the SBC and the service delivery rates for these customers will include Efficiency Transition Implementation Plan ("ETIP") costs.

Q. DID THE RDP SPECIFY WHICH ENERGY EFFICIENCY PROGRAMS WOULD BE AVAILABLE TO TC/IT CUSTOMERS?

A. No. However, the City issued discovery to the Companies seeking more information on this topic (please refer to Exhibit ___(NYCRDP-3), which is National Grid's response to CNY-53). This interrogatory sought information with regard to which energy efficiency programs would be offered to Non-firm customers, when they would be eligible, and the steps that National Grid has taken to assess Non-firm customer interest in the programs, as well as any outreach efforts undertaken by National Grid to attract Non-firm customers to its programs. National Grid's response listed several programs "that would possibly be offered to KEDNY and KEDLI Non-firm customers." The Companies "did not propose a specific start date for Non-firm customers to participate in energy efficiency programs" but stated that they are not opposed to a Rate Year start date. Finally, the Companies stated that they "have not undertaken steps to assess Non-firm customer interest in participating in energy efficiency programs, or planned outreach to attract Non-firm customers."

Q. BASED ON THE RDP'S PROPOSAL AND NATIONAL GRID'S RESPONSE TO CNY-53, WHAT IS YOUR RECOMMENDATION REGARDING NON-FIRM CUSTOMERS PARTICIPATING IN AND BEING CHARGED FOR ENERGY EFFICIENCY PROGRAMS?

A. The Companies' proposal to apply the SBC to Non-firm customers and that ETIP costs be included in their service delivery rates should not be adopted at this time. The Companies' response to CNY-53 indicates that they are not prepared to offer energy efficiency programs to Non-firm customers yet. Therefore, the Companies should not be allowed to charge Non-firm customers for programs that are not being specifically offered to them.

Having said this, based on my conversations with City of New York personnel, the City supports allowing Non-firm customers to participate in, and pay for, energy efficiency programs that are targeted to their needs. If the Companies would respond with programs in which the City's Non-firm accounts could participate, as well as commit to outreach to the City and other Non-firm customers explaining the benefits of energy efficiency programs, then we likely would support charging the SBC and ETIP to Non-firm customers.

Q. HOW SHOULD THE SBC BE APPLIED TO THE TIER 1 AND TIER 2 RATES OF THE NON-FIRM CUSTOMERS, ASSUMING THE COMPANIES OFFER SPECIFIC ENERGY EFFICIENCY PROGRAMS FOR SUCH CUSTOMERS?

A. The total Tier 1 and Tier 2 volumetric rates should not increase over the currently proposed Tier 1 and 2 rates because these rates are already greatly in excess of cost-based rates for Non-firm customers. This means that the if the SBC is added to these customers' rates, there should be a corresponding reduction in the applicable volumetric rates.

Q. WILL THE NEW SC 22 BE SUBJECT TO THE COMPANIES' EARNINGS ADJUSTMENT MECHANISM ("EAM")?

A. No, SC 22 will not be subject to the EAM. National Grid confirmed this in its response to CNY-58, which we have included in Exhibit ___(NYCRDP-4). Currently, TC/IT customers are not subject to the Companies' EAM and it should be made clear that the EAM will not apply to the new TC/IT tariffs proposed by the Companies.

Q. DO YOU HAVE ANY FURTHER RECOMMENDATIONS TO THE COMMISSION?

A. Yes. We recommend that the Demand Response Demonstration Project ("DRD Project") be expanded beyond what the Companies propose.

Q. PLEASE EXPLAIN WHY YOU ARE MAKING THIS RECOMMENDATION TO THE COMMISSION.

A. The Companies' Future of Heat ("FOH") Panel acknowledged that demand response "has the potential to play an important role in the Companies' future of heat strategy." Direct Testimony of the Future of Heat Panel, page 41, lines 7 through 8. The FOH Panel cited two ways in which the Companies are promoting demand response. The first way is the Companies' Non-firm service offerings, which we address earlier and again below. The second is the DRD Project for firm service customers who commit to minimum reductions

when called upon to do so. The FOH Panel presented the very positive results of the DRD Project on pages 43 and 44 of its Direct Testimony. Exhibit ___(FOH-1), Schedule 5 presents the Companies' proposed expansion of the DRD Project.

Q. HAS THE CITY PARTICIPATED IN THE DRD PROJECT?

A. Yes. The City is an active participant in the DRD Project and, in general, is pleased with how the program is run.

Q. WHAT ROLE SHOULD DEMAND RESPONSE PLAY?

A. As noted by the NYC Policy Panel, demand response should play an increased role in managing peak loads, particularly given prevailing capacity constraints. As reported by the FOH Panel, the DRD Project has proven successful and, as a result, the Panel is proposing to expand the program to achieve 1,000 Dt/hour and 400 Dt/hour reductions for KEDNY and KEDLI, respectively, by FY 2024 (Exhibit___FOH-1, Schedule 5).

Q. CAN YOU COMMENT ON THE TARGETS OF THE EXPANDED DRD PROJECT?

A. Yes. Given its success and the increasing need for peak load reductions, the Companies should increase the targets of the DRD Project. This could be achieved by expanding the pool of eligible customers, reducing the requirement for a 10 percent or more committed reduction, increased customer recruitment, or through other measures. Given that the Companies have declared a gas moratorium, expansion of the DRD Project should be explored in a collaborative that includes interested stakeholders.

Q. DO YOU HAVE ANY OTHER RECOMMENDATIONS REGARDING NON-FIRM SERVICE OPTIONS?

A. Yes. We recommend expanding Non-firm service eligibility to customers who do not have an alternate fuel source but have backup facilities, are willing to shed load automatically and be remotely interrupted by the Companies, and otherwise comply with the Companies' tariffs. The Non-firm tariffs currently require that a customer **must** have alternate fuel on site to be eligible. It is possible that customers could provide interruptions on demand (i.e., by automatic switching) without alternate fuels. For example, NYCHA is exploring a boiler design that would allow for interruptions on demand by starting up equipment that does not require natural gas or fuel oil. The Companies should be able to accommodate this configuration by providing an electronic or other remote signal to a Non-firm customer that triggers automatic shut-off of a portion or all of the onsite Non-firm gas service, whether or not that shut-off is accompanied by a transition to fuel oil. The primary (and perhaps only) concern from National Grid's standpoint should be whether the gas usage is interrupted at the customer's premises. To the extent the Companies have technological limitations that prevent automatic shutoff of Non-firm gas service in the absence of fuel oil backup, those limitations also need to be addressed and removed to ensure that Non-firm service options are not unnecessarily restrictive. Indeed, expanding the eligibility criteria so that on-site alternate fuel capability is not a prerequisite may spur other innovative ways to provide Non-firm service, without waiving the performance standards otherwise applied to such service.

Q. HOW SHOULD THE EXPANSION OF ELIGIBILITY FOR NON-FIRM SERVICE PROCEED?

A. Expansion of the pool of Non-firm customers could be included in the collaborative we recommended earlier. That collaborative also could address inspection and other rights the Companies may need to ensure that the customer equipment is capable of receiving interruption signals and responding to them, similar to the rights the Companies have to ensure that dual-fuel customers are capable of interrupting when called upon.

Q. WHAT OTHER RATE DESIGN ISSUES DOES THE PANEL WANT TO RAISE?

A. On August 11, 2019, the City submitted Comments in response to the Companies' May 8, 2019 tariff filings ("Filings"). The Filings were made in compliance with the Commission's TC/IT Order issued on February 7, 2019 in Cases 16-G-0058 and 16-G-0059. The City's Comments are attached hereto as Exhibit ___(NYCRDP-5).

In the Comments, the City noted its general support for the consolidation of the TC and IT service classes, as well as the new Non-firm rates set forth in the Filings. However, as noted in the Comments, the new tariffs should be amended to allow Non-Firm customers with automatic-switching equipment to elect the Tier 2 delivery rate, at their option.

The City's Comments also requested that the Commission require that the Companies eliminate the "two-strike" rule, which dictates that customers who do not interrupt twice in any winter period either be transferred to firm service or, in the alternative, have their gas service terminated. In addition, the City's Comments also recommended that all Non-firm customers have the option to remain on that service, subject to paying penalties and filing a remedial plan with the Companies, and that the current criteria for remaining on Non-firm service after "two strikes" be modified. Finally, the Comments requested that the tariff definition of critical care customers be expanded.

To the extent that the Commission determines that changing the "two-strike" rule or the recommendations regarding critical care customers, as set forth in the City Comments, are outside the scope of the Filings, the City requests, for the reasons set forth in Exhibit ___(NYCRDP-5), that such relief be granted in these proceedings.

Q. DOES THE CITY HAVE ANY FURTHER COMMENTS ON THE TARIFFS ADDRESSING NON-FIRM SERVICE?

A. Yes. Given the recent gas moratoriums declared by the Companies, the Companies and interested parties should be looking for ways to incentivize customers to apply for, or stay on, Non-firm service.

Q. PLEASE CONTINUE.

A. Earlier in this testimony, we urge the Commission to require the Companies to commence a collaborative to expand Non-Firm service so that it is not limited exclusively to customers who have on-site alternate fuel capability. Expanding Non-firm service will allow the Companies to more effectively manage their systems in the face of dwindling capacity.

Q. ARE THERE OTHER CHANGES THAT WOULD IMPROVE THE NON-FIRM SERVICE OPTION?

A. Yes. First, the requirement for a second mandatory test is not necessary and is costly to customers, and should be eliminated. Second, the monetary penalty for so-called "second strike" violations should be reduced or waived, and the violation should not count as a "strike," if the second interruption notice is issued within five days of the end of a prior event and the customer can demonstrate that it could not repair the equipment that prevented it from interrupting during the first event. Finally, under certain circumstances

monetary penalties otherwise applicable to the NYCHA for operating on-site generation facilities to provide electricity to human needs customers during electric emergencies should be waived.

Q. PLEASE EXPLAIN WHY THE MANDATORY SECOND TEST SHOULD BE ELIMINATED.

A. The addition of a second, unannounced test at the end of January each year commenced in 2017. The additional test adds costs to customers, increases emissions and creates uncertainty. In addition, it is not clear why it is necessary, particularly for customers with equipment for automatic-switching controlled by the Companies. The Companies typically test their TC/IT customers at the beginning of each winter to ensure that they can respond. Once that is established, the tariff imposes numerous obligations on the TC/IT customer to maintain adequate alternate fuel supplies and to keep any necessary dual-fuel equipment operational. In fact, the Companies have the ability to inspect and test such alternate fuel facilities at any time, and the customer has to grant access to the Companies for inspection and testing purposes. Finally, as discussed in more detail below, there are significant penalties associated with non-compliance, and this alone provides a tremendous incentive for maintaining alternate fuel facilities.

Q. WHAT IS THE RATIONALE FOR REDUCING THE LEVEL OF THE PENALTY UNDER CERTAIN CIRCUMSTANCES?

A. The City appreciates the prior revision that changes the unauthorized use penalty from the higher to the **lesser** of (i) two times the sum of the Market Price for natural gas delivered to the Company's city gate on the day of violation plus the transportation rate the Customer

would be subject to if it were a transportation Customer, or (ii) nine times the applicable sales rate, using nine times the applicable sales rate is excessive. However, as evidenced by the amnesty clause included in the Companies' tariffs, there may be circumstances where the failure to interrupt is not due to a conscious act but, instead, equipment failure. For example, if the first "strike" is due to equipment failure, and despite diligent efforts to fix the problem, the equipment may not be able to be fixed prior to a second event, especially if the second event is in close proximity after the first event. In this case, we would request a threshold of within five days of the end of the first event to qualify as "close proximity." The rationale for this threshold is that the inability to fix the equipment could be due to long lead times or transportation problems, in particular if there is a weekend that falls between the end of the first and the beginning of the second interruption event. If that is the case, the imposition of the penalty is not providing any incentive to change behavior. Accordingly, and as exemplified in concept by the amnesty clause, fairness would dictate that the penalties for the second "strike" should be significantly reduced or waived, and the event should not be considered a "strike." The City acknowledges that the waiver of second strike penalty should require submission of documentation to show a good faith attempt to fix the equipment.

Q. DO YOU HAVE ANY ADDITIONAL RECOMMENDATIONS REGARDING THE APPLICATION OF PENALTIES?

A. Yes. NYCHA currently has or will install natural gas-fired back-up emergency generators on site at 32 of its housing developments in the wake of Hurricane Sandy. Of these, 15 developments are located in National Grid's service territory. This on-site generation is

designed to provide critical electric service to almost 10,000 apartments that house approximately 25,000 of the City's neediest residents during emergencies that interrupt electric service. Although NYCHA requested firm service for these installations, the moratorium has cast doubt on whether they will be able to get that service. Given the summer peaking nature of Con Edison's electric system, outages are far more likely to occur in the summer months rather than the winter months when electricity usage is lower. However, there may be rare instances in which the affected NYCHA apartments experience electric outages during peak gas usage periods in the winter. If the NYCHA facilities with back-up generation are relegated to interruptible gas service and experience an electric outage at the same time that National Grid calls an interruption event for TC/IT customers, then NYCHA would have to make the very difficult choice of either not running its backup generators for the health and safety of its residential customers during an emergency or incurring significant and costly penalties from National Grid. This "lose-lose" situation for NYCHA is the result of the Companies declaring gas moratoriums, and it is not fair that NYCHA residents, including those that suffered significant deleterious impacts during Hurricane Sandy, might have to bear the brunt of decisions wholly outside of their control. Accordingly, penalties imposed on NYCHA should be waived under the unique circumstances set forth here. To address the broader issue of the need for improved comprehensive supply planning, please refer to the City's Policy Panel, which has a number of specific recommendations addressing the gas moratorium.

Q. WHAT IS YOUR RECOMMENDATION?

A. As noted in this testimony, the City is not opposed to the Companies applying reasonable penalties during periods in which interruptions have been called to deter customers from using gas so that they can safely manage their systems. However, in the rare instance that Con Edison experiences an electric outage and NYCHA has to run its back-up generators to protect the health and safety of its human needs customers during a period when National Grid has called for an interruption, it would be appropriate to waive the penalties in this limited circumstance, especially considering that the NYCHA facilities requested firm service. The City is willing to have appropriate NYCHA personnel to sit down with National Grid to establish workable protocols, including establishing the amount of gas needed to maintain critical electric service.

Q. DOES THIS CONCLUDE YOUR DIRECT TESTIMONY?

A. Yes.